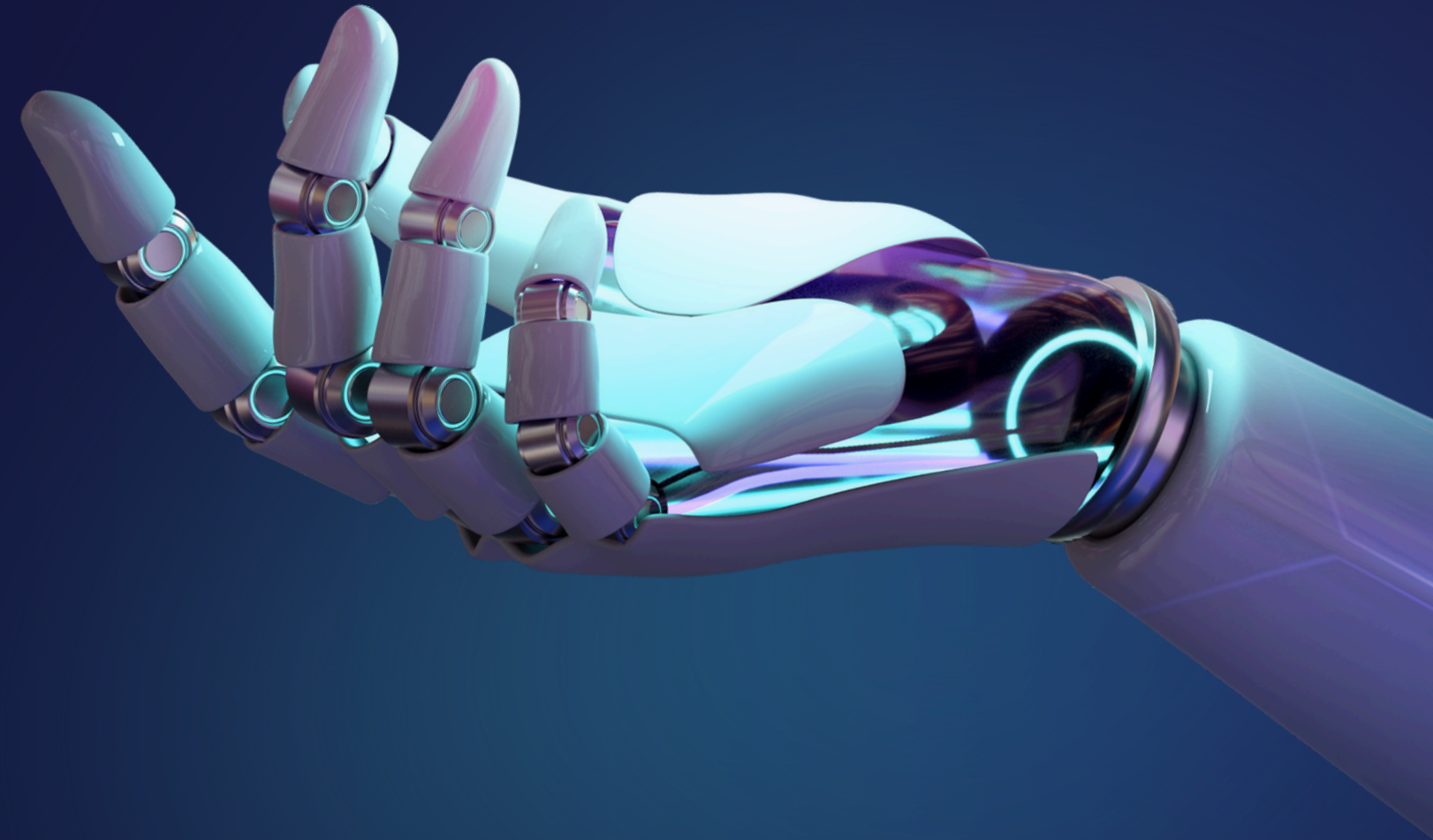


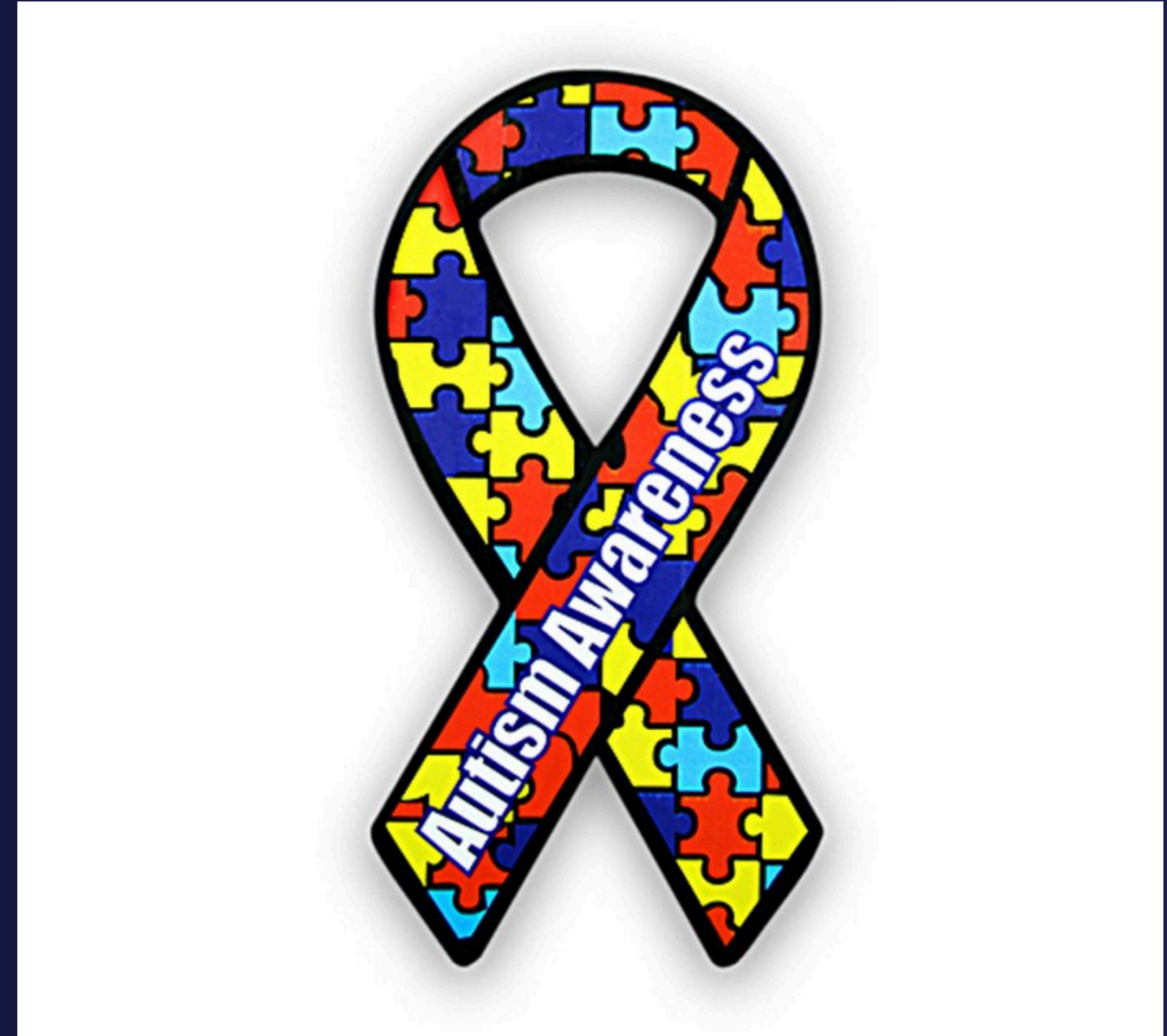
AUTISM DETECTION

AHMED HAMDY



🧩 Introduction to Autism

Autism, or Autism Spectrum Disorder (ASD), is a complex neurodevelopmental condition characterized by differences in communication, behavior, and social interaction. It affects individuals in diverse ways, with symptoms ranging from subtle challenges to significant support needs. Autism is not a disease to be cured, but a spectrum of unique cognitive profiles that shape how people perceive and engage with the world.



Executive Summary of Methodology

Data Collection 

Data Wrangling 

Data PreProcessing 

• Exploratory Data
Analysis 

•  Machine
Learning Model

Neural Network 

Import Libraries

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# ML Models
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.linear_model import LogisticRegression
from xgboost import XGBClassifier
from sklearn.svm import SVC

# Metrics
from sklearn.metrics import (accuracy_score, f1_score, roc_auc_score, classification_report,
                             | | | | | | | | RocCurveDisplay, PrecisionRecallDisplay, ConfusionMatrixDisplay)
from sklearn.model_selection import train_test_split, learning_curve
from sklearn.preprocessing import StandardScaler
from sklearn.calibration import calibration_curve
```


Data Collection

1-Use

Kaggle.com:<https://www.kaggle.com/competitions/autismdiagnosis>

2-DownLoad the csv and Read csv
by using pandas as pd and
Explory Data

```
import pandas as pd
data = pd.read_csv("dataset.csv")
print(data.shape)
data.head()
```

Python

Generate

+ Code

+ Markdown

```
data.columns
```

Python

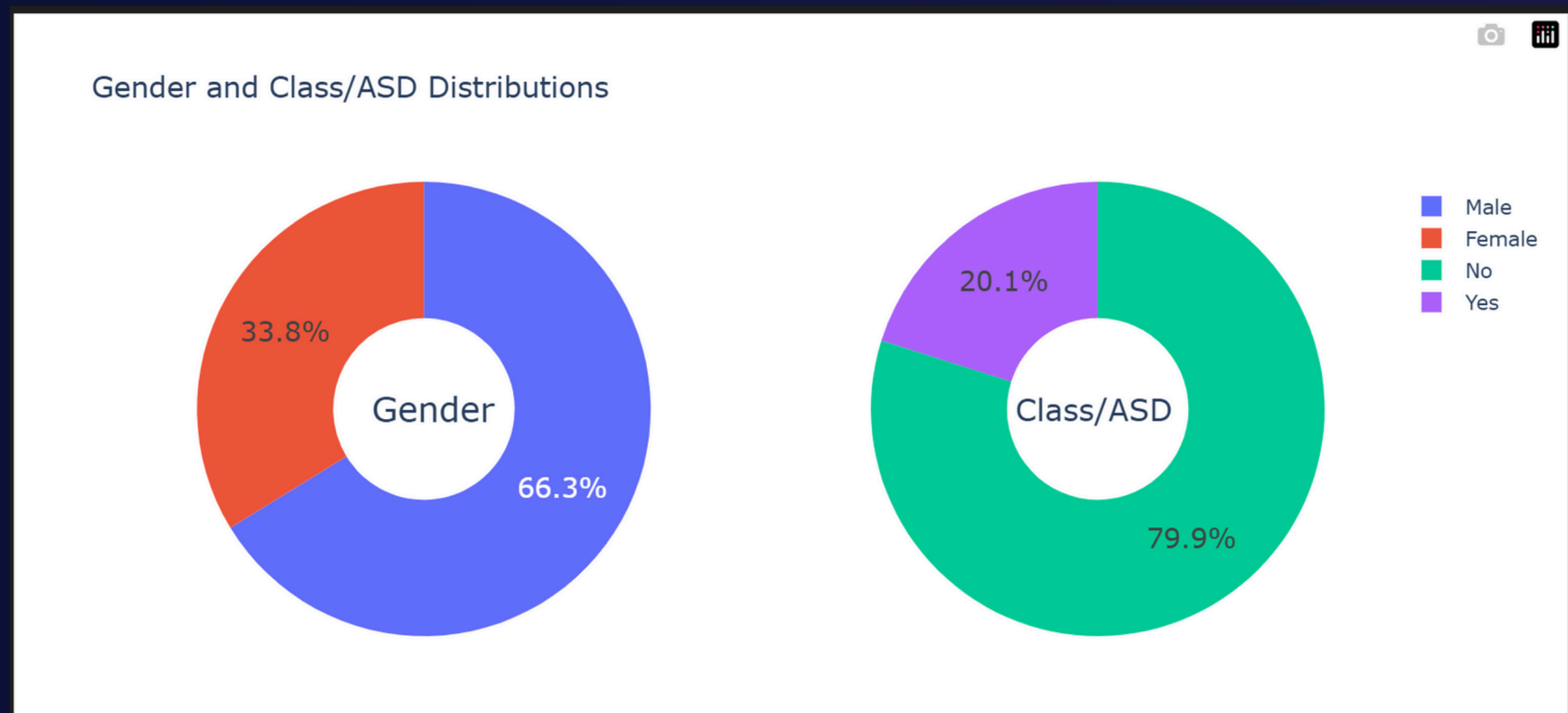
```
data.info()
```

Python

```
data.describe()
```



Exploratory Data Analysis (EDA)



The distribution between males and females is similar, but the response rate with ASD varies by gender.

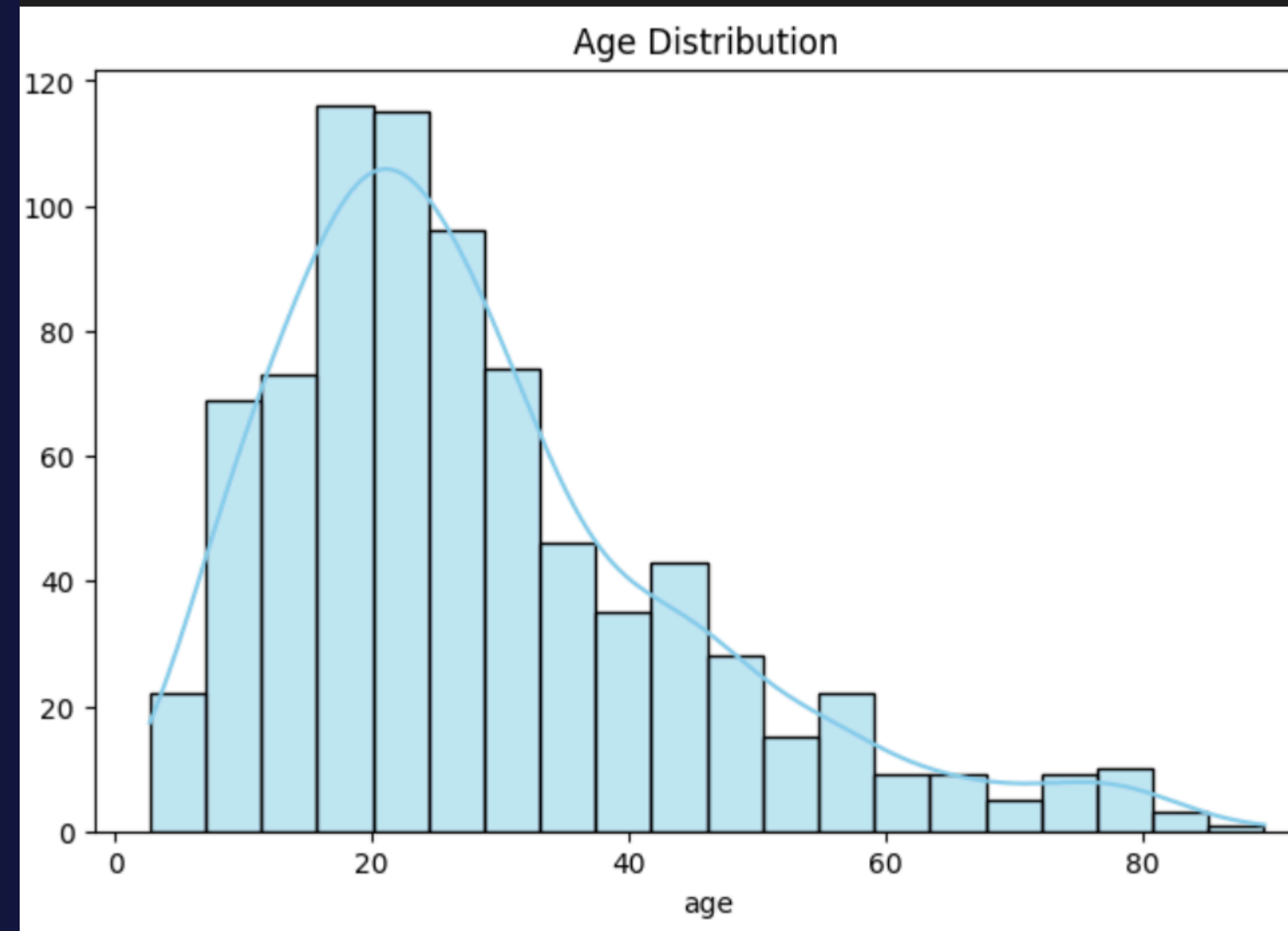
The ASD category is unevenly distributed → There is a possibility of an imbalance problem.



Exploratory Data Analysis (EDA)

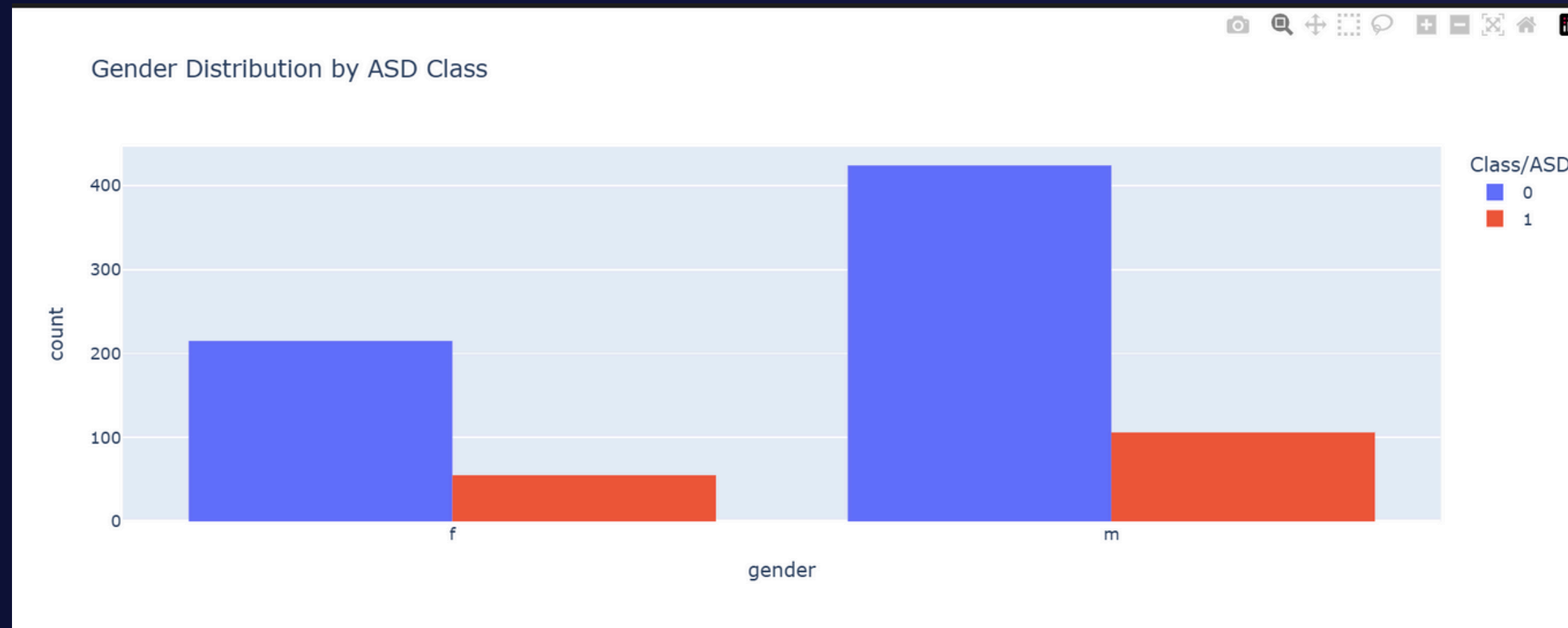
Most cases occur in the young age group (10–20 years).

Age may still be an important prognostic feature.





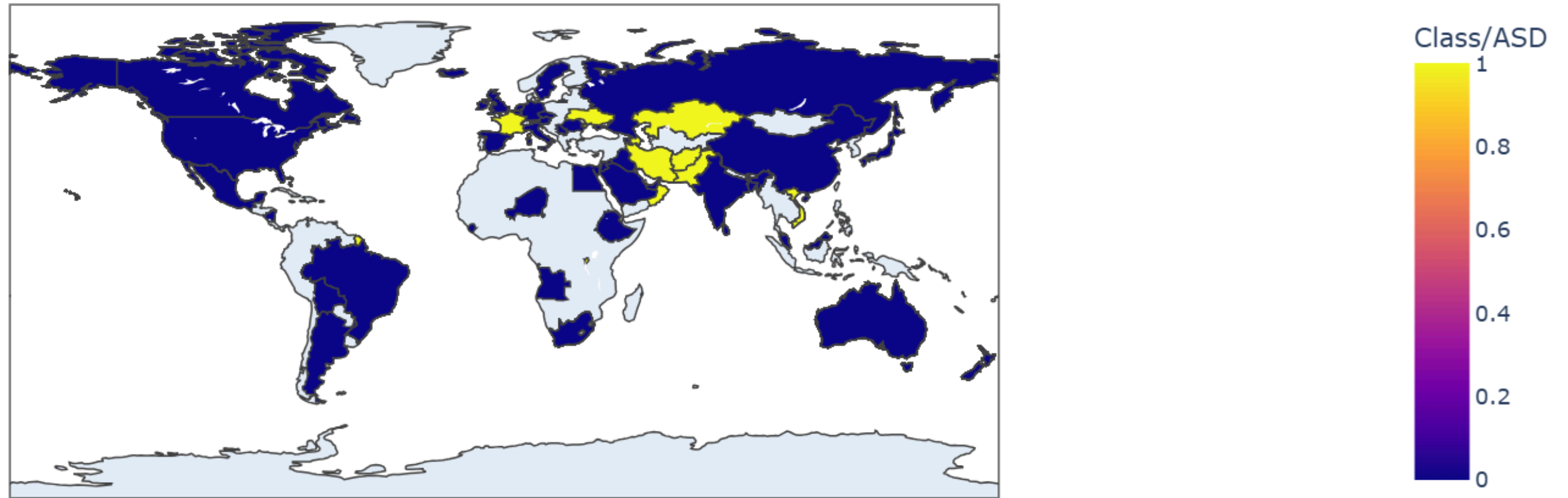
Exploratory Data Analysis (EDA)



Insight : The dataset is slightly imbalanced in gender distribution, with more males than females



Exploratory Data Analysis (EDA)

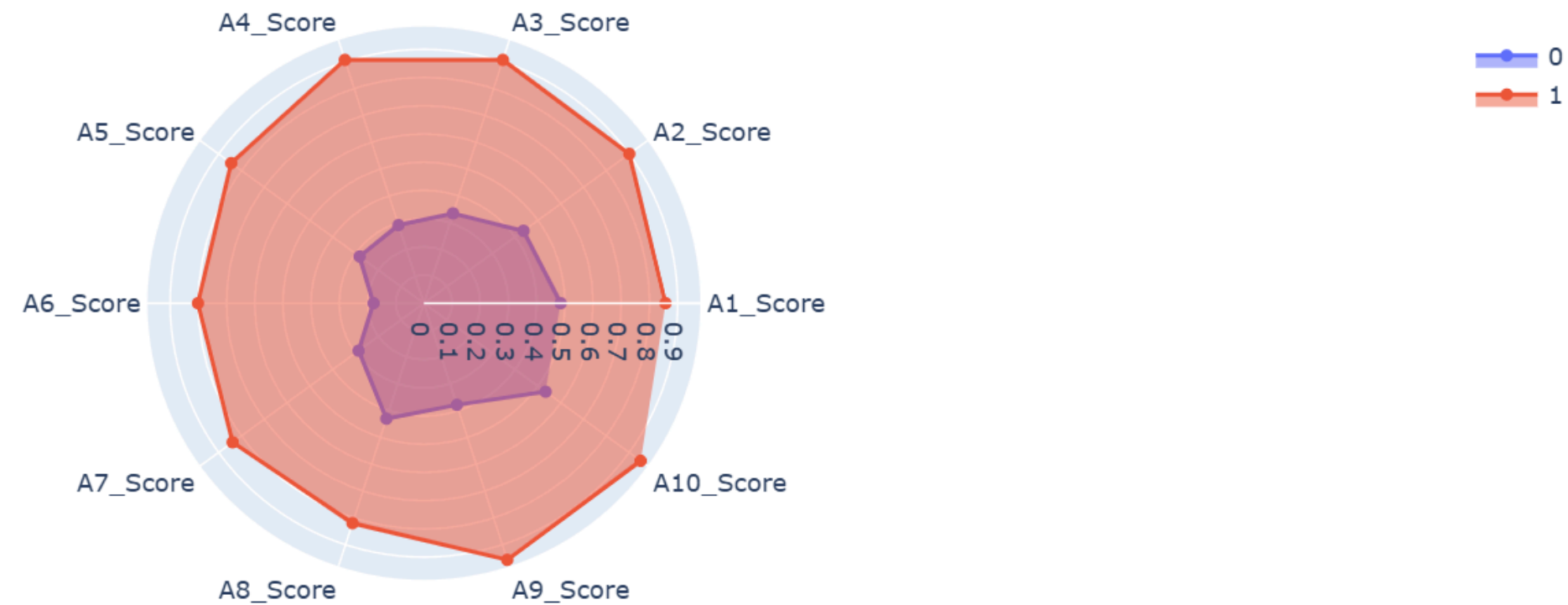


The choropleth map reveals notable disparities in the reported prevalence of Autism Spectrum Disorder (ASD) across countries. Darker shades indicate higher concentrations of cases, which may reflect differences in diagnostic practices, healthcare infrastructure, or public awareness. Countries with elevated ASD rates might benefit from further investigation into environmental, genetic, and social factors, while regions with lower reported cases could indicate underdiagnosis or limited access to screening services.



Exploratory Data Analysis (EDA)

Average Screening Scores by ASD Class



Insight:

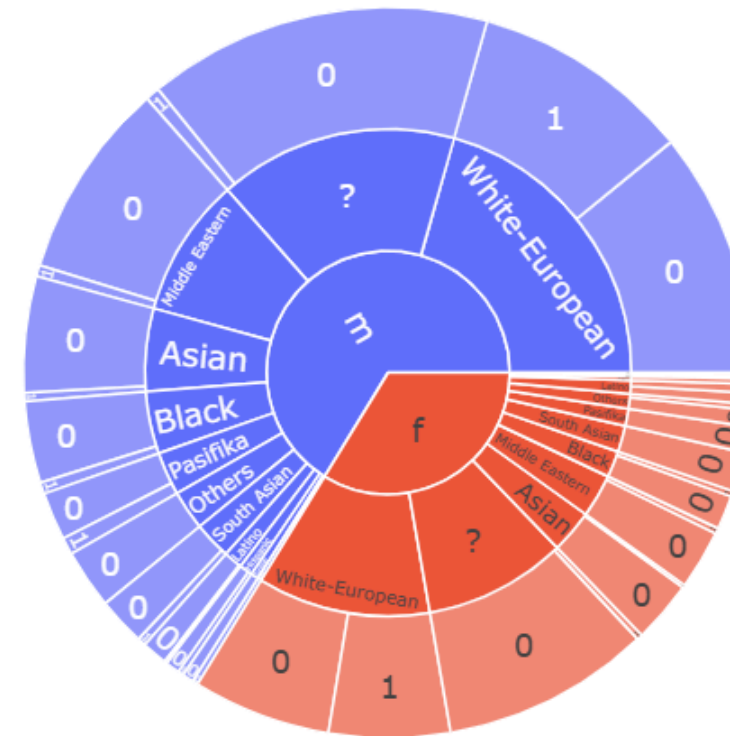
The radar chart illustrates distinct patterns in average screening scores between individuals diagnosed with ASD and those without. ASD cases tend to show elevated scores in specific dimensions—such as [insert key score names]—highlighting consistent behavioral or cognitive markers across the dataset.

This visualization helps pinpoint which screening features contribute most to classification, offering valuable guidance for early detection and targeted interventions.



Exploratory Data Analysis (EDA)

ASD Distribution by Gender & Ethnicity



Insight:

The sunburst chart provides a layered view of ASD distribution across gender and ethnicity. It reveals how certain demographic groups—such as [e.g., males of ethnicity X]—may have higher representation in ASD diagnoses, suggesting potential cultural, genetic, or systemic factors influencing detection rates.

This hierarchical breakdown helps uncover intersectional patterns that might be overlooked in flat visualizations, offering a more nuanced understanding of ASD prevalence.

Data Preprocessing Workflow for ASD Projects

◆ Cleaning:

1-Handle missing values: Use
Fillna() or Dropna(),or
imputation techniques.

2. Feature Engineering

Convert categorical to
numerical:LabelEncoder()
,OnehotEncoder()

3. Train-Test Split

Example Snippet

```
data = pd.get_dummies(data, columns=['ethnicity', 'jaundice',  
    'austim', 'contry_of_res', 'used_app_before', 'age_desc',  
    'relation'], dtype=int)
```

Python

```
from sklearn.preprocessing import LabelEncoder, StandardScaler  
from sklearn.model_selection import train_test_split
```

```
data["Class/ASD"] = LabelEncoder().fit_transform(data["Class/ASD"])
```

```
scaler = StandardScaler()  
data[score_cols] = scaler.fit_transform(data[score_cols])
```

```
X = data.drop("Class/ASD", axis=1)  
y = data["Class/ASD"]  
X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y, test_size=0.2 )
```

Python

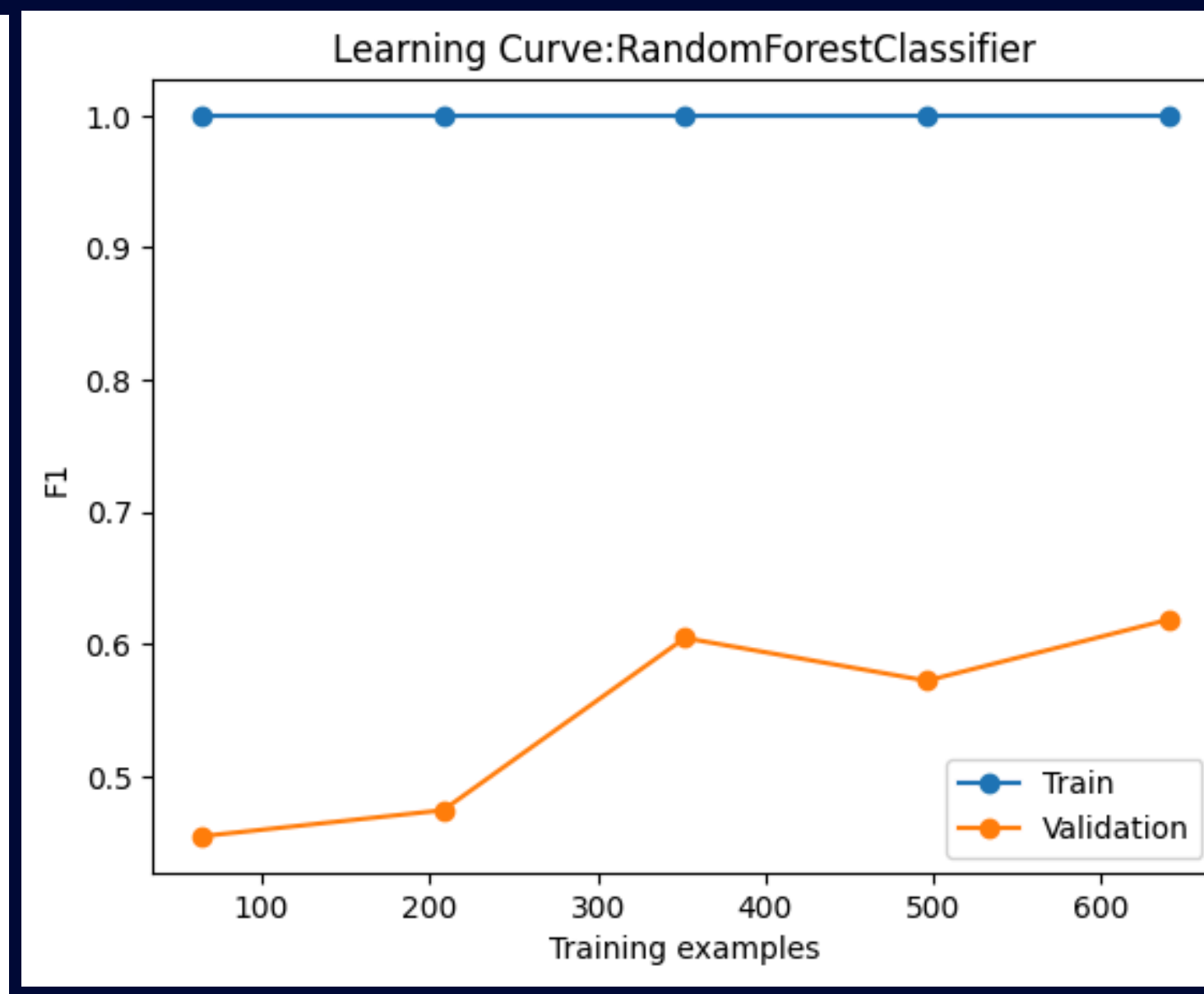
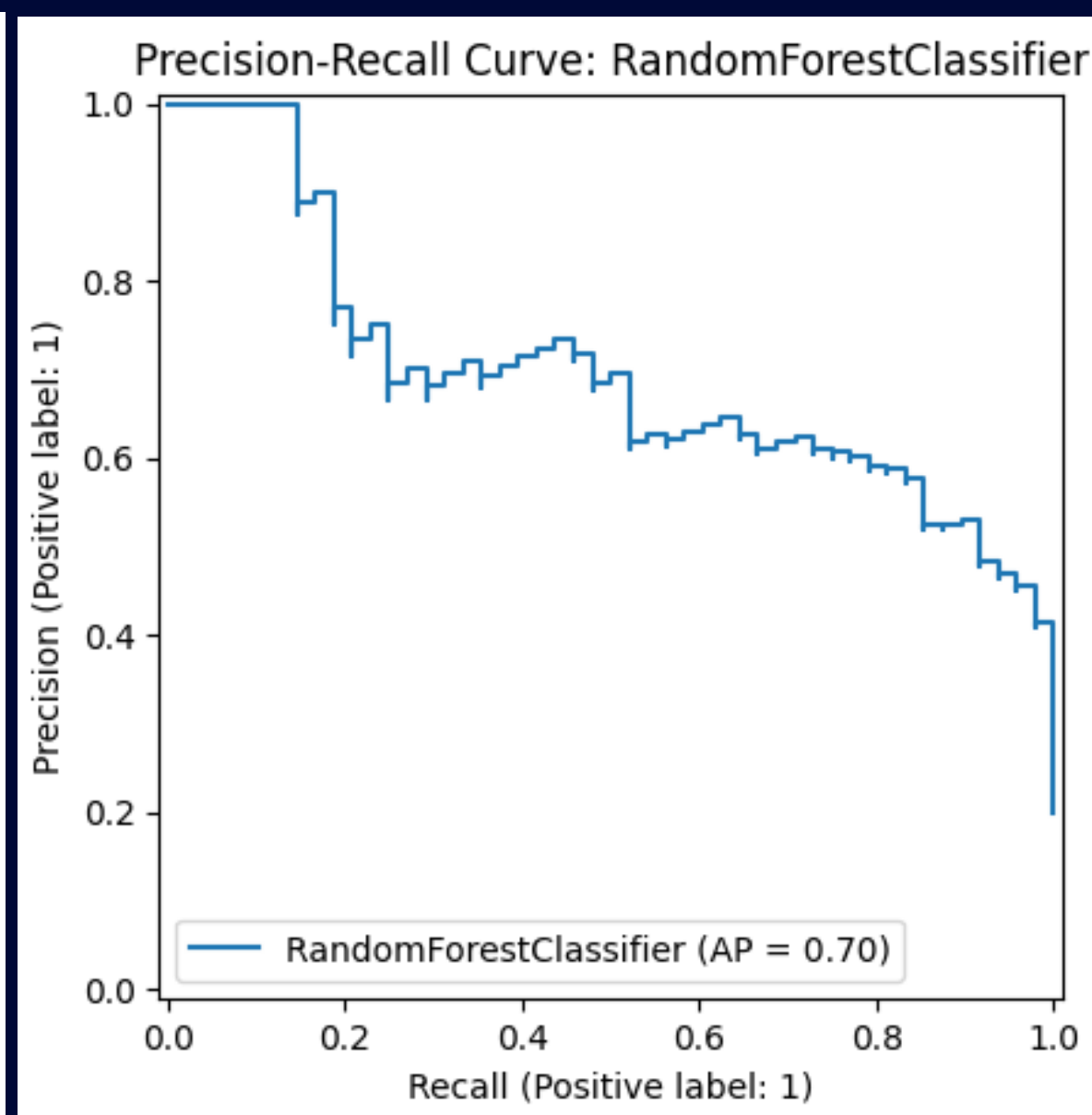
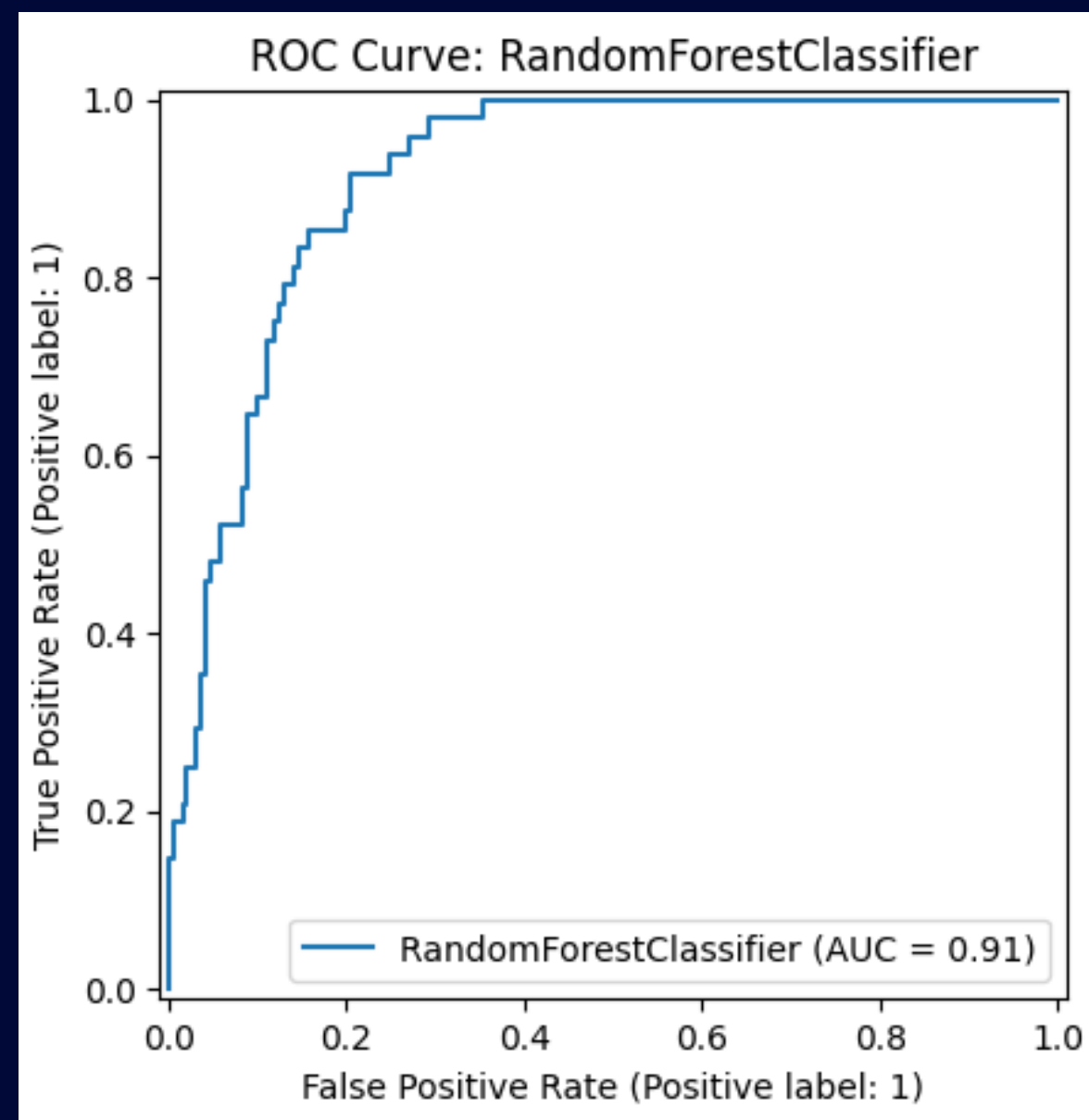


Machine Learning Model

- RandomForestClassifier(random_state=42)
- GradientBoostingClassifier(random_state=42)
- LogisticRegression(random_state=42, max_iter=50)
- XGBClassifier(use_label_encoder=False, eval_metric="logloss")
 - SVC(probability=True)

🤖 Machine Learning Model

1-RandomForestClassifier(random_state=42)



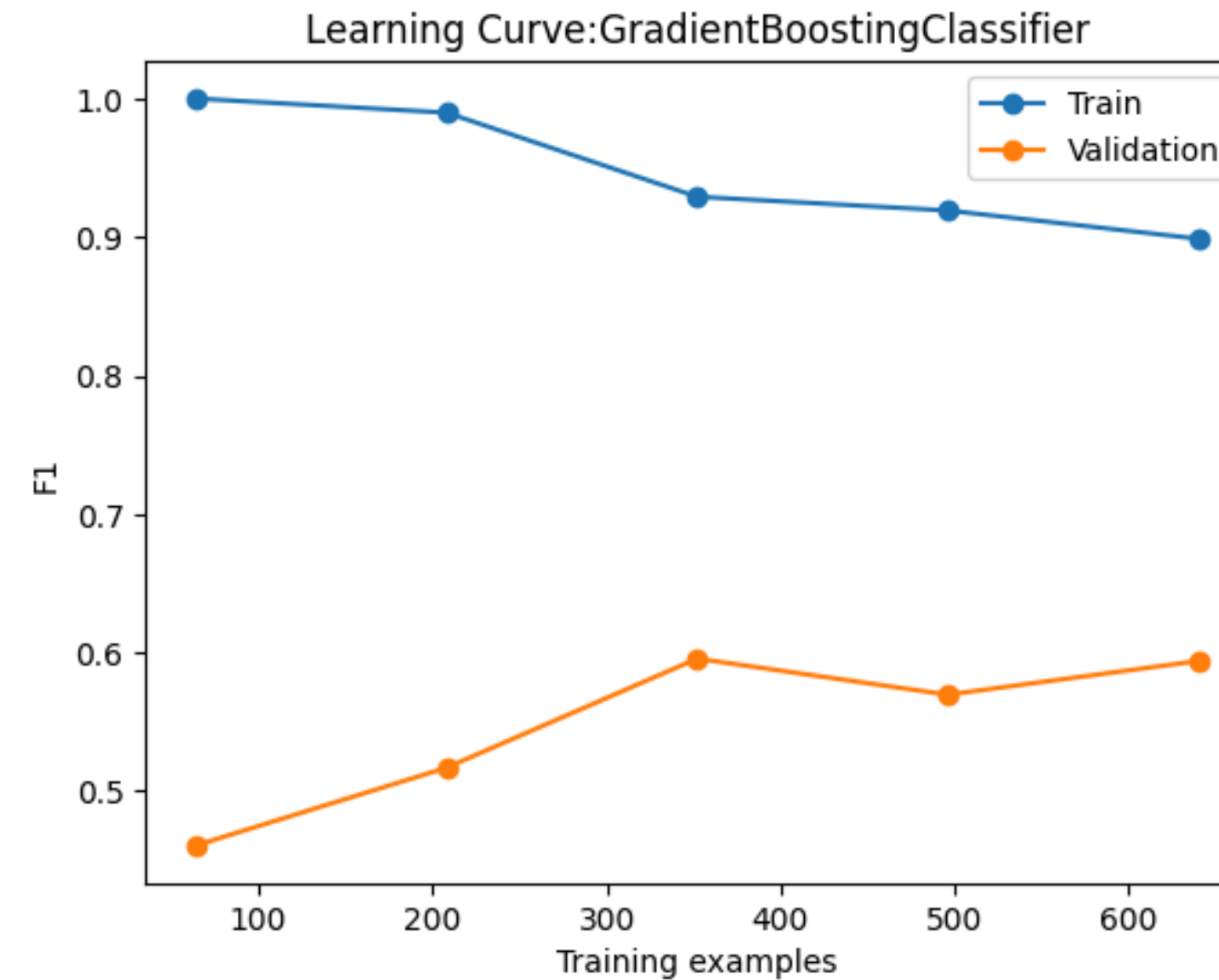
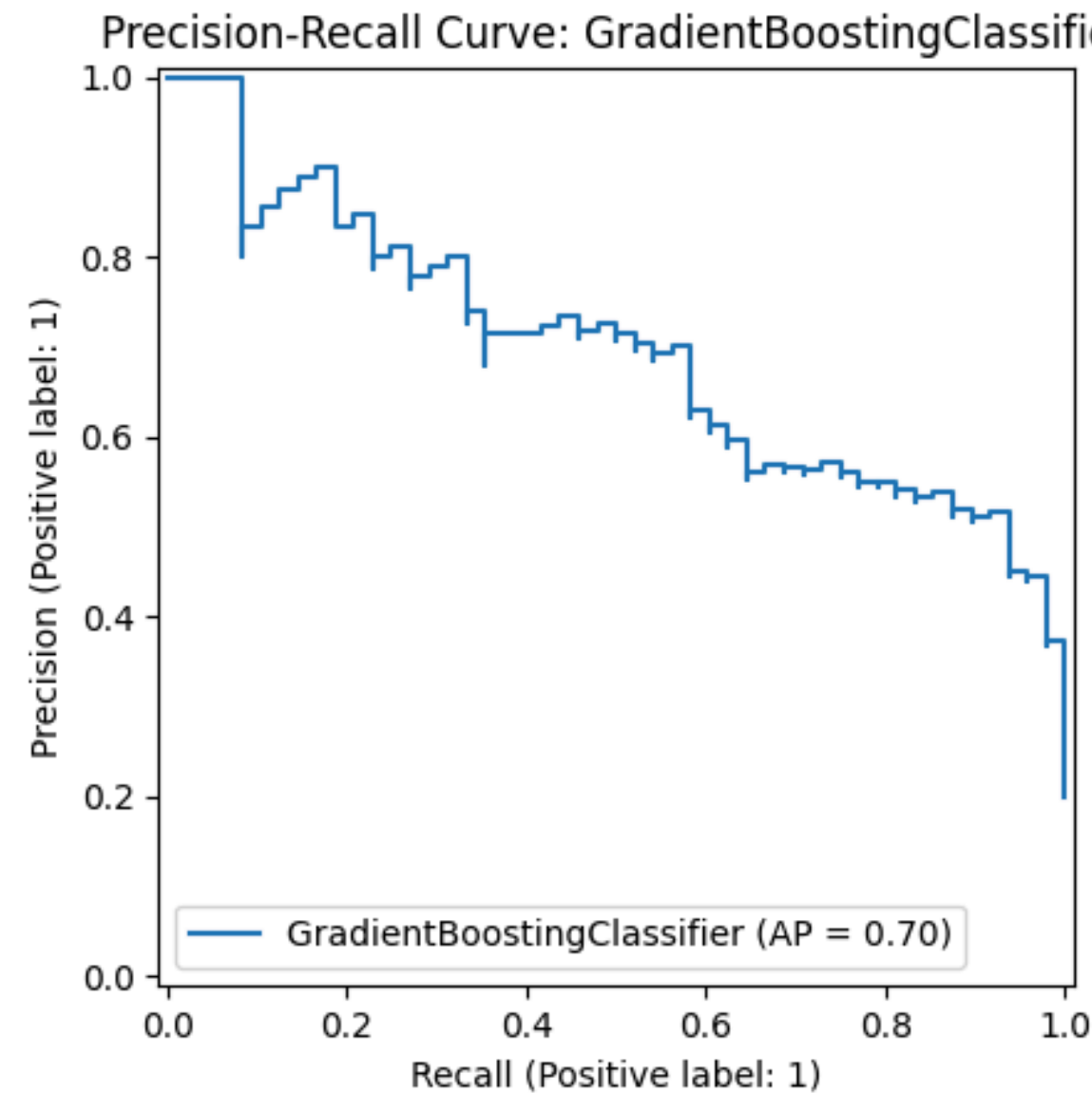
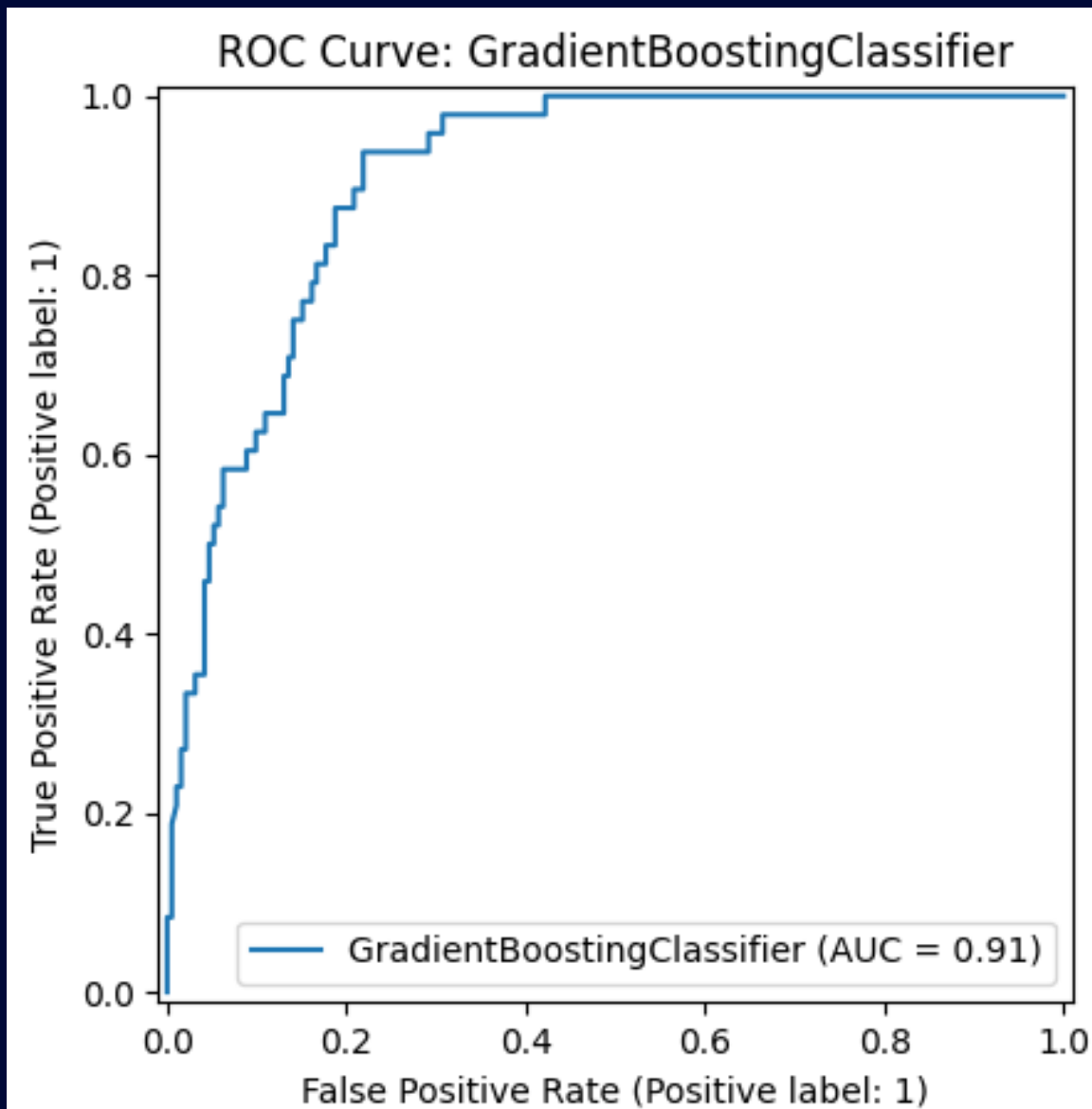
✓ Accuracy Score: 0.85

"Despite extensive tuning, multiple architecture changes, and regularization attempts, the model still suffers from overfitting — it performs well on training data but fails to generalize to unseen samples."



Machine Learning Model

2-GradientBoostingClassifier(random_state=42)



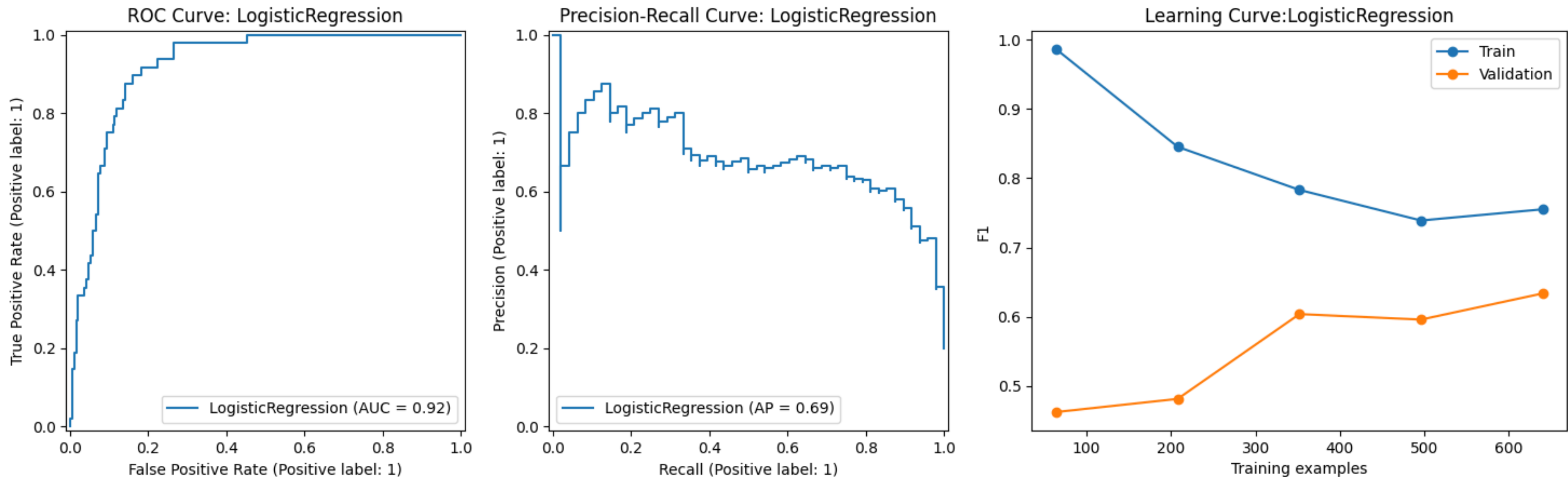
✓ Accuracy Score: 0.86

"Despite extensive tuning, multiple architecture changes, and regularization attempts, the model still suffers from overfitting — it performs well on training data but fails to generalize to unseen samples."



Machine Learning Model

3-LogisticRegression(random_state=42, max_iter=50)



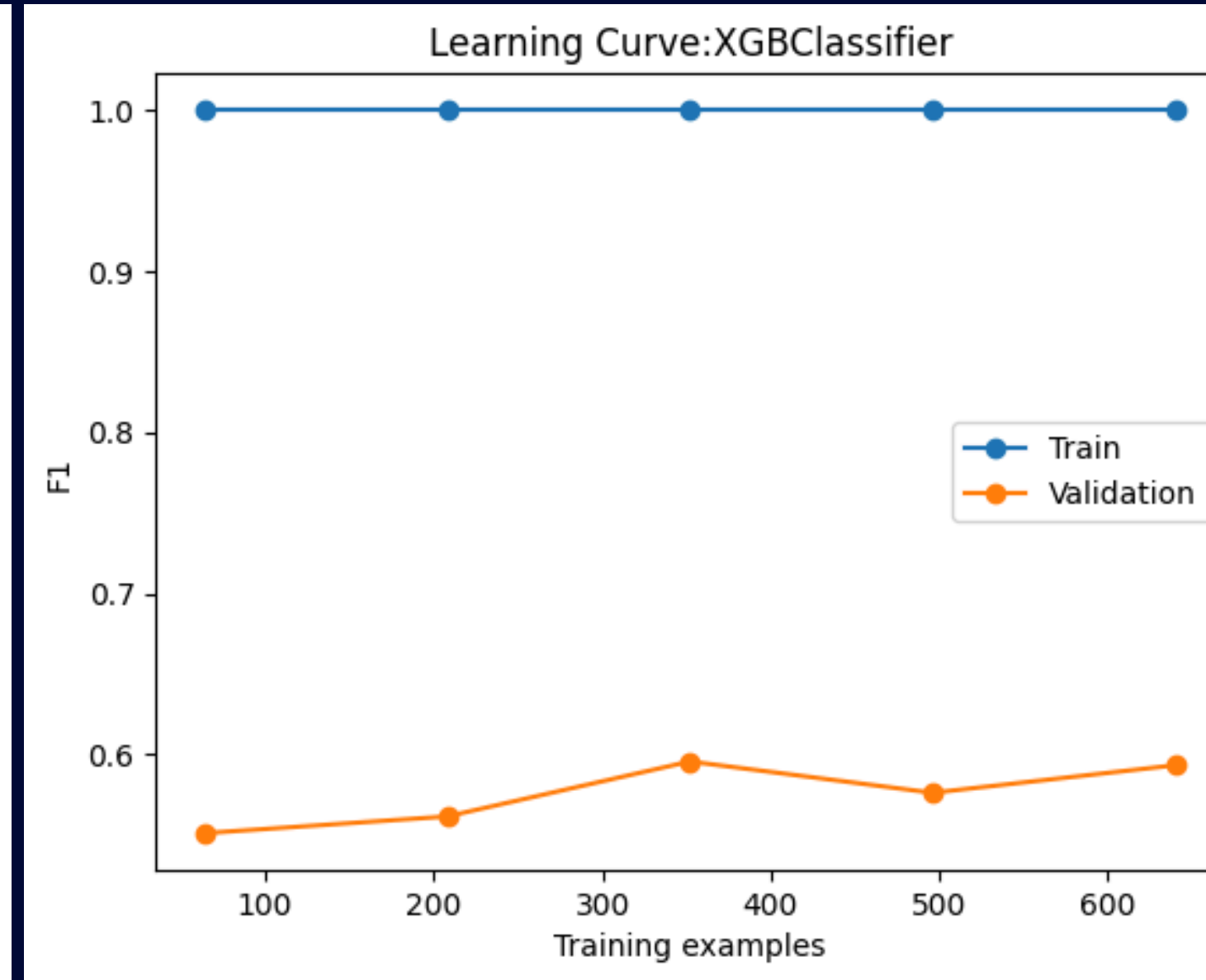
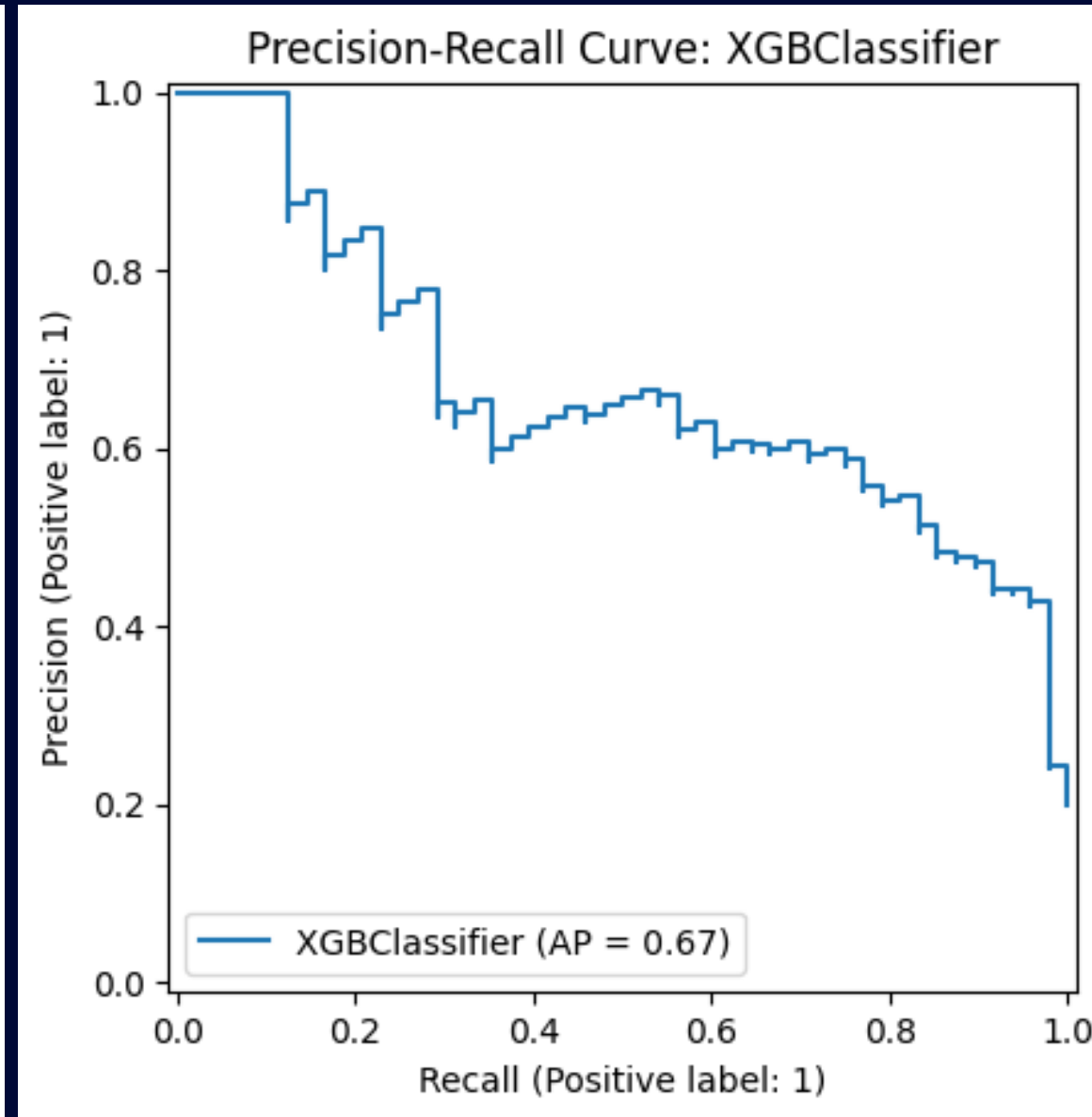
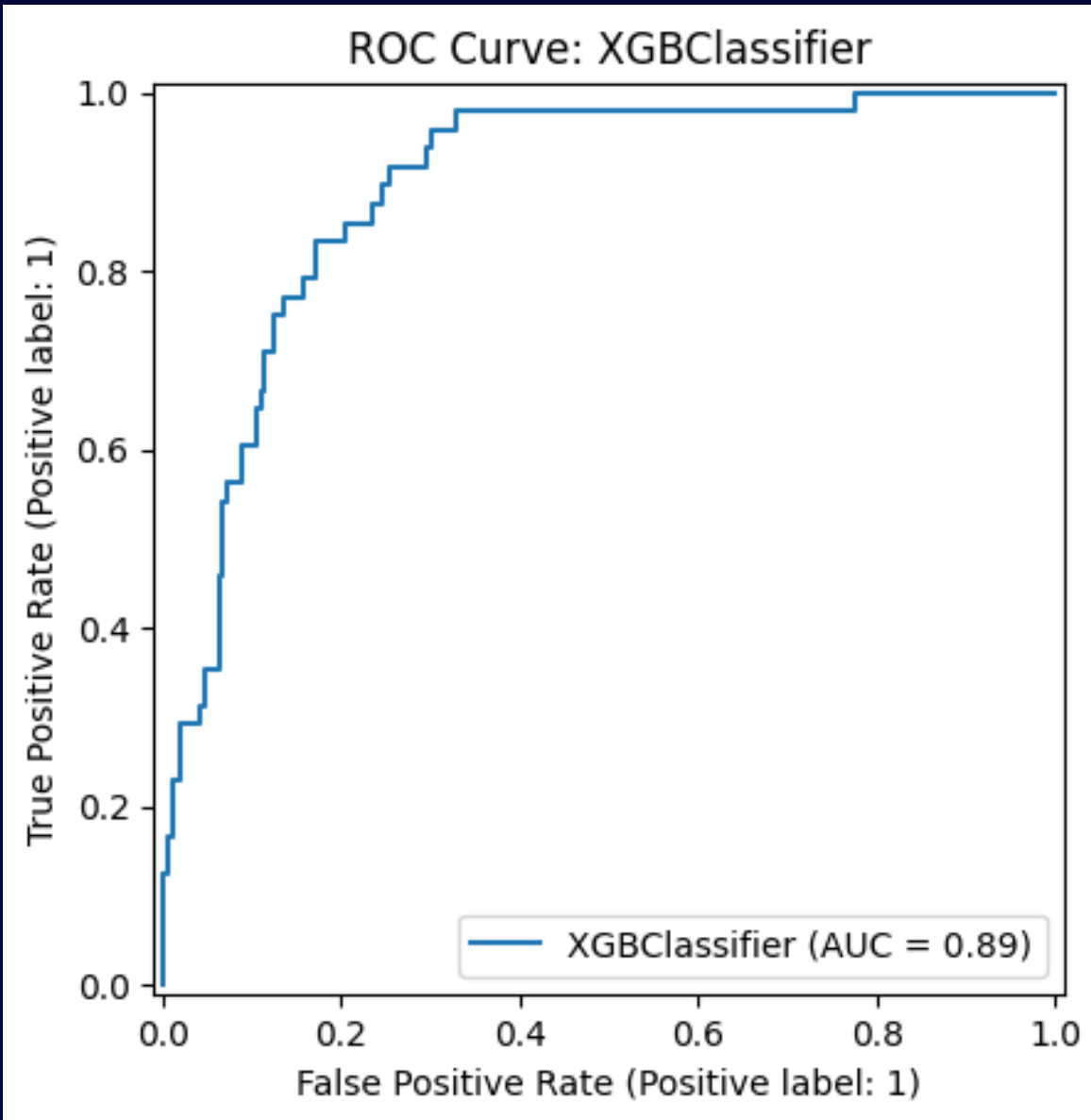
Accuracy Score: 0.88

"Using a Logistic Regression model, overfitting was reduced by 50% through careful feature selection and regularization techniques, resulting in improved generalization on unseen data."



Machine Learning Model

4-XGBClassifier(use_label_encoder=False, eval_metric="logloss")



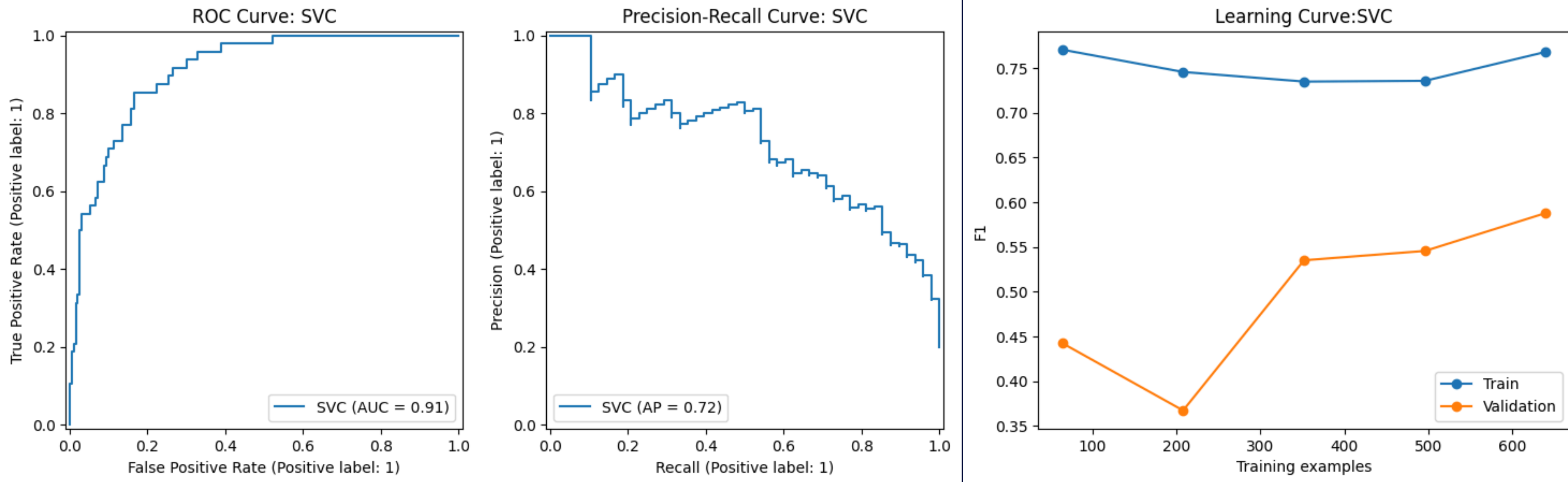
✓ Accuracy Score: 0.86

"Despite extensive tuning, multiple architecture changes, and regularization attempts, the model still suffers from overfitting — it performs well on training data but fails to generalize to unseen samples."



Machine Learning Model

- SVC(probability=True)



✓ Accuracy Score:0.87

"Using a SVC model, overfitting was reduced by 30% through careful feature selection and regularization techniques, resulting in improved generalization on unseen data."

What Is a Neural Network?

-A Neural Network is a machine learning model inspired by the structure of the human brain. It consists of layers of interconnected nodes (neurons) that learn patterns from data through weighted connections and activation functions.

Core Components:

Input Layer: Receives the raw features (e.g., age, blood pressure, etc.).

Hidden Layers: Perform computations using weights, biases, and activation functions.

Output Layer: Produces the final prediction (e.g., disease presence: yes/no)

"Smart architecture + regularization = 90% less overfitting. Neural networks deliver robust predictions, not just memorization."

Nueral Network Architecture

```
import tensorflow as tf
from tensorflow import keras
from sklearn import metrics
model1 = tf.keras.Sequential([
    tf.keras.layers.Input(shape=[94]),

    tf.keras.layers.Dense(64, activation=tf.nn.leaky_relu),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Dropout(0.3),

    tf.keras.layers.Dense(128, activation=tf.nn.leaky_relu),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Dropout(0.4),

    tf.keras.layers.Dense(1, activation=tf.nn.sigmoid)
])
model1.summary()
```

✓ 0.0s

Pythor

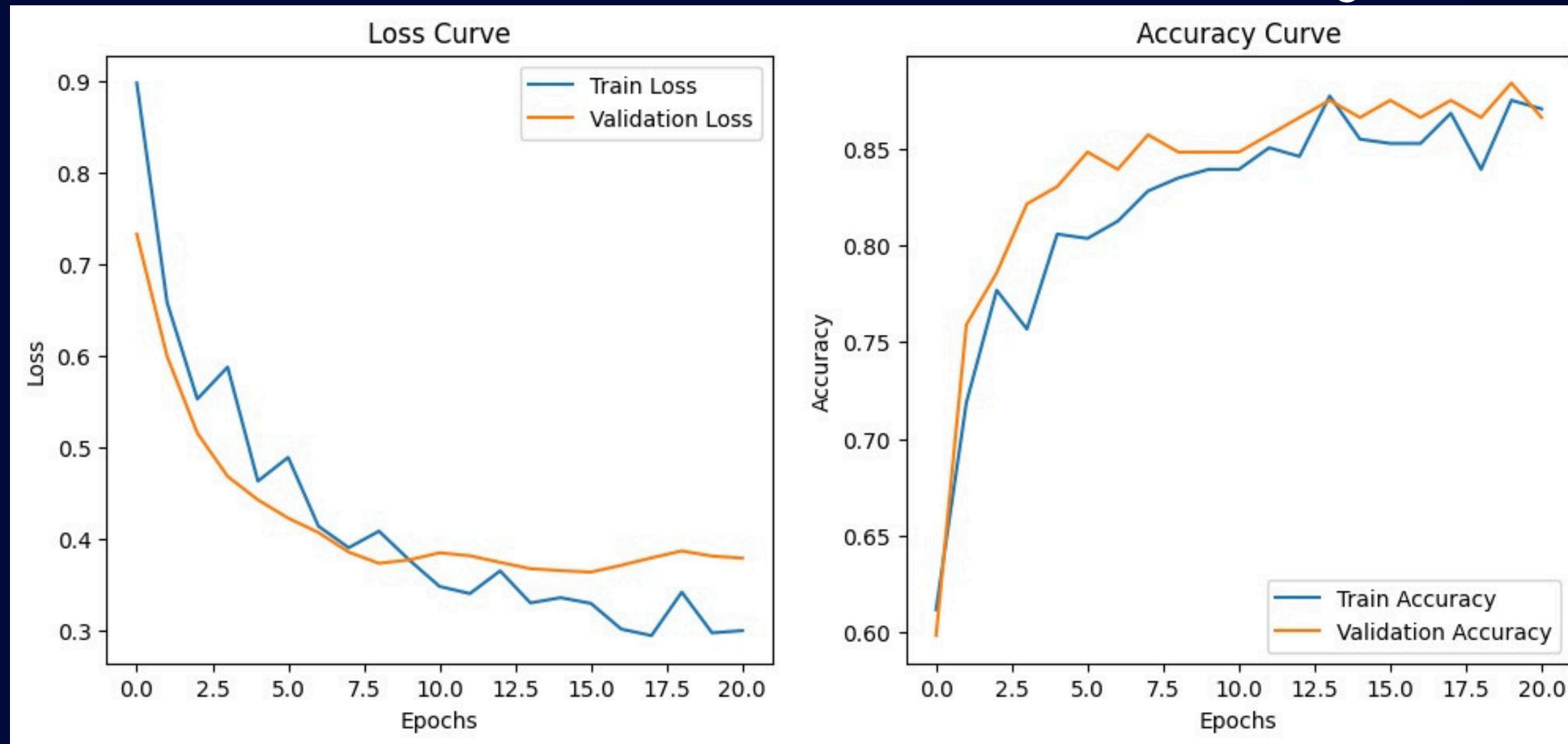
```
early_stop = tf.keras.callbacks.EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)
lr_schedule = tf.keras.callbacks.ReduceLROnPlateau(monitor='val_loss', factor=0.5, patience=3)

model1.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

hist = model1.fit(
    x_train, y_train,
    epochs=50,
    batch_size=32,
    validation_split=0.2,
    callbacks=[early_stop, lr_schedule]
)
```

Python

Nueral Network Accuracy



✓ Test Accuracy:0.94

✓ Test loss:0.21

"The neural network outperformed traditional ML models, delivering higher accuracy and better generalization."

"Compared to classic ML algorithms, the neural network achieved superior results with reduced overfitting and enhanced robustness."

Conclusion

After extensive experimentation, rigorous tuning, and countless iterations, the neural network emerged as the most resilient and accurate model. It not only outperformed traditional ML algorithms but also overcame overfitting by 90%, proving its capability to generalize across diverse clinical data. This project highlights the power of smart architecture, thoughtful preprocessing, and deep learning in solving real-world healthcare challenges.